

Notes: Fisman & Wang (RES, 2015)

The Mortality Cost of Political Connections

Contents

1	Big picture and incremental contribution	3
1.1	Incremental contribution of the paper	3
2	Data and measurement	4
2.1	Sample and industries	4
2.2	Worker-fatality measures	4
2.3	Political connections	4
2.4	Safety violations and pollution violations	4
2.5	NSNP laws and regulator incentives	5
2.6	Firm controls and outcomes	5
3	Theoretical framework and empirical specifications	5
3.1	Two-period safety-compliance model	5
3.2	Main fatality regressions	5
3.3	Regulatory enforcement regressions	6
3.4	Profit and market-reaction regressions	6
3.5	NSNP interaction regression	6
4	Identification strategy	6
4.1	Core empirical challenge	6
4.2	Within-firm identification	7
4.3	Timing of connections	7
4.4	Testing the mechanism	7
4.5	Alternative explanations addressed	7
5	Tables and figures	8
5.1	Figure 1: Frequency distribution of deaths	8
5.2	Tables 1A and 1B: Summary statistics and connected versus unconnected firms	8
5.3	Table 1C and Figures 2A–2B: Within-firm variation and industry death rates	9
5.4	Tables 2 and 3: Political connections and workplace deaths	10
5.5	Tables 4 and 5: Safety violations and environmental non-compliance	11
5.6	Table 6 and Figure 3: Firm performance and market reaction to fatal accidents	11
5.7	Tables 7 and 8: Event-study regressions and loss of connections	12
5.8	Tables 9 and 10: NSNP laws and regulator incentives	13
5.9	Table 11 and Appendix Tables A.1–A.4: Robustness and institutional details	13

6	Interpretive synthesis	15
7	Short summary	15
8	Conclusion	16
8.1	Core conclusion	16
8.2	Economic intuition	16
8.3	Incremental contribution in one sentence	16
8.4	Takeaway	16

1 Big picture and incremental contribution

- **Question.** Do political connections allow firms to evade socially valuable safety regulation, and does that evasion impose measurable mortality costs on workers?
- **Setting.** The paper studies publicly listed Chinese firms from 2008–2013 in sectors subject to production-safety regulation, combining executive political-background data with worker fatality records.
- **Main empirical fact.** Politically connected firms have worker death rates roughly two to three times higher than comparable unconnected firms, depending on the sample and specification.
- **Mechanism.** Connected firms appear to be monitored less intensely before fatal accidents: in no-fatality firm-years, connected firms are much less likely to have public reports of safety violations.
- **Post-accident evidence.** Fatal accidents produce negative abnormal stock returns for connected firms, connected firms' profit premium disappears after fatal accidents, and connected executives are more likely to leave after fatal accidents.
- **Regulatory-incentive evidence.** Provincial NSNP laws that tied safety regulators' promotion prospects to safety targets substantially reduce the excess death rate at connected firms.
- **Core interpretation.** The evidence fits a model in which connections allow firms to circumvent costly safety compliance ex ante, but connections lose value after fatal accidents attract scrutiny.
- **Policy message.** Political ties can have direct social costs, but changing regulator incentives can mitigate those costs.

1.1 Incremental contribution of the paper

- **Moves the political-connections literature from private rents to social costs.** Earlier work often measures whether connected firms obtain higher profits, market value, credit, contracts, or subsidies. This paper asks whether those private gains come at the cost of worker lives.
- **Creates a firm-level worker-fatality panel for publicly listed Chinese firms.** The paper combines official records, company reports, CSR reports, construction-accident data, and media searches to measure worker deaths by firm-year.
- **Links executive biographies to regulatory evasion.** The connectedness measure is based on C-suite executives' prior government office, allowing the paper to connect individual political backgrounds to firm safety outcomes.
- **Separates ex ante compliance avoidance from ex post protection.** The paper studies safety violations in no-fatality years, violations after fatal accidents, stock-market reactions, profit changes, and executive departures. This combination distinguishes two mechanisms: avoiding compliance before accidents versus avoiding punishment afterward.
- **Introduces regulator incentives as a mitigating institution.** By exploiting provincial NSNP reforms that made safety officials' promotion depend on safety targets, the paper shows that institutional incentives can weaken the mortality cost of connections.
- **Broadens corporate social responsibility evidence.** It shows that governance and political economy can matter for hard social outcomes—worker deaths—not only accounting outcomes, ESG disclosures, or reported compliance.

2 Data and measurement

2.1 Sample and industries

The empirical sample consists of Chinese publicly listed firms from 2008 through 2013 in industries where worker-safety regulation is salient and fatalities are observable. The industries include coal mining, chemicals, ferrous and nonferrous metals, construction, and other safety-relevant industries. The final firm-year panel contains 1,475 observations and 1,332 worker deaths.

Interpretation: The paper focuses on publicly listed firms because executive biographies, financial controls, and accident-related information can be collected systematically. The downside is that these firms may be more scrutinized than private firms, so the results need not measure the average mortality cost for all Chinese firms.

2.2 Worker-fatality measures

Worker deaths are collected from official sources, company reports, CSR reports, construction-industry accident data, and WiseSearch media searches. For construction firms, accident-level data come from the Ministry of Housing and Urban-Rural Development. For missing cases, media searches supplement official and corporate sources.

The paper uses four main death measures:

$$Deaths_{cy}, \quad I(Deaths_{cy} > 0), \quad DeathRate_{cy} = 1000 \times \frac{Deaths_{cy}}{Employment_{cy}}, \quad Severe_{cy} = 1[\text{at least 3 fatalities in one acc}]$$

DeathRate is winsorized at the 1 percent level. A fatality occurs in 13.5 percent of firm-year observations.

Interpretation: Deaths are harder to hide than injuries and are therefore less gameable. Severe accidents are especially useful for robustness because official and company records are more consistent for accidents with at least three fatalities.

2.3 Political connections

A firm-year is coded as connected if, at the beginning of the year, a senior executive—chairman, vice-chairman, CEO, or vice-CEO—previously held a sufficiently high government position. The relevant threshold is mayor or vice-mayor in the city where the firm is located, or a provincial or central government official of the same or higher rank. More than 90 percent of connected executives in the sample came from municipal bureaucracy.

Interpretation: Timing matters. Connections are measured before the accident year, which reduces reverse causality from deaths to connectedness and helps the firm fixed-effects specifications use executive turnover as identifying variation.

2.4 Safety violations and pollution violations

Firms are not required to report safety violations. The paper constructs SafetyViol using media reports about major government investigations for workplace safety violations. Pollution fines are collected as an additional proxy for regulatory noncompliance and enforcement.

Interpretation: Safety-violation reports are imperfect because they require public disclosure. But they are useful precisely because lower violation reporting at connected firms despite higher fatalities suggests reduced oversight rather than better safety.

2.5 NSNP laws and regulator incentives

The paper studies provincial NSNP policies that made promotion of safety regulators and local officials contingent on meeting safety targets. Provinces adopted these reforms at different times. The paper uses province-year variation in whether a province had passed an NSNP law in a previous year.

Interpretation: NSNP reforms are not randomized, but they shift regulator incentives in a way that should especially matter for connected firms if their advantage comes from avoiding monitoring.

2.6 Firm controls and outcomes

Control variables include log sales, return on assets, capital intensity, state ownership, investment, and firm employment. The paper also studies market reactions through cumulative abnormal returns (CARs), and firm performance through current and future ROA.

Interpretation: The combination of fatality outcomes, enforcement outcomes, profitability, stock returns, and executive turnover lets the paper triangulate the mechanism rather than relying on the fatality regression alone.

3 Theoretical framework and empirical specifications

3.1 Two-period safety-compliance model

The paper motivates the empirical analysis with a simple model. Firms choose whether to hire a connected executive and whether to shirk on safety expenditures. Connected executives take a share α of the profit gains from avoiding safety expenditures. A fatal accident occurs with probability p when the firm shirks, causes a fine F , and destroys the value of the connection.

Expected profits can be summarized as:

$$E(\pi_i) = \begin{cases} 2(R - C - s_i), & \text{Connected}_i = 0, \\ (1 - \alpha)(R - C) + p(R - C - s_i - F) + (1 - p)[(1 - \alpha)(R - C) - pF], & \text{Connected}_i = 1. \end{cases}$$

A firm hires a connected executive when

$$s_i \geq (R - C)\alpha + pF.$$

Interpretation: The model predicts three main facts: connected firms shirk more on safety, connected firms are more profitable when no fatal accident occurs, and fatal accidents reduce the value of connections.

3.2 Main fatality regressions

The main regression relates death outcomes to political connections:

$$Y_{cy} = \beta \text{Connected}_{cy} + X'_{cy}\gamma + \lambda_y + \lambda_j + \lambda_p + \epsilon_{cy},$$

where Y_{cy} is either *Deaths*, $I(\text{Deaths} > 0)$, or *DeathRate*, and fixed effects vary across specifications. Some specifications include firm fixed effects, so the estimate is identified by within-firm changes in connections due to connected-executive arrival or departure.

Interpretation: The coefficient on *Connected* is not a fully causal effect because executives are not randomly assigned. The firm-fixed-effect versions are therefore important because they remove time-invariant firm risk differences.

3.3 Regulatory enforcement regressions

For safety and environmental enforcement, the dependent variables are *SafetyViol* and *Pollute*. The key question is whether connected firms are less likely to be investigated or fined, especially in years without fatal accidents.

Interpretation: A lower enforcement rate at connected firms despite higher fatality rates is hard to reconcile with better safety. It is consistent with avoidance of monitoring before fatalities become visible.

3.4 Profit and market-reaction regressions

The paper estimates effects of political connections on ROA and next-year ROA, and examines abnormal stock returns around fatal accidents:

$$CAR_{c,[1,K]} = \beta Connected_c + X'_c \gamma + \epsilon_c.$$

It also estimates whether fatal accidents predict loss of connections:

$$Connected_{cy} = \beta_0 + \beta_1 I(Deaths_{c,y-1} > 0) + \beta_2 Connected_{c,y-1} + X'_{c,y-1} \gamma + \epsilon_{cy}.$$

Interpretation: These regressions ask whether the value of political connections disappears after a fatal accident. If connections are valuable because they help avoid safety compliance, a public accident should make the connection less useful and should lower firm value.

3.5 NSNP interaction regression

To test regulator incentives, the paper estimates models such as:

$$Y_{cy} = \beta_1 Connected_{cy} + \beta_2 NSNP_{py} + \beta_3 Connected_{cy} \times NSNP_{py} + X'_{cy} \gamma + FE + \epsilon_{cy}.$$

A negative β_3 implies that regulator-incentive reforms reduce the excess deaths associated with connected firms.

Interpretation: This is the main institutional test. If safety officials face career penalties for deaths, political connections should become less useful as a shield from monitoring.

4 Identification strategy

4.1 Core empirical challenge

Political connections are not randomly assigned. Connected firms may differ from unconnected firms in size, industries, risk, profitability, or managerial ability. The paper therefore uses multiple complementary strategies rather than relying on a single causal design.

4.2 Within-firm identification

The firm fixed-effects specifications use changes in Connected within the same firm, arising from the arrival or departure of connected executives. This removes fixed firm differences in technology, safety culture, local geography, or industry risk.

Interpretation: This is the strongest part of the death regressions because the connected-unconnected comparison is made within the same firm over time.

4.3 Timing of connections

Connected is measured at the beginning of the calendar year, before fatalities in that year. This timing makes it more plausible that connectedness precedes deaths rather than deaths attracting connected executives.

4.4 Testing the mechanism

The paper tests several mechanism predictions:

- connected firms have higher death rates;
- connected firms are less often publicly reported for safety violations in no-fatality years;
- connected firms are less often fined for pollution;
- connected firms are more profitable before accidents;
- fatal accidents generate negative abnormal returns at connected firms;
- connected executives are more likely to depart after fatal accidents;
- NSNP reforms reduce connected firms' excess death rates.

Interpretation: No single test proves causality. The identification argument is cumulative: all of these facts are easiest to reconcile with ex ante avoidance of costly compliance.

4.5 Alternative explanations addressed

- **Under-reporting:** severe accidents are used because deaths are harder to hide; Table 11 finds connected firms also have more severe accidents.
- **Industry mix:** industry fixed effects and industry-specific trends are included in robustness tests.
- **Size and state ownership:** log sales, employment, capital intensity, investment, ROA, and state ownership are controlled.
- **Managerial talent or growth motives:** connected firms do not simply appear to be expanding employment or sales to meet political goals.
- **Regulatory backlash avoidance:** violation rates after fatal accidents are similar for connected and unconnected firms, suggesting connections do not mainly protect firms after a visible accident.

Interpretation: The paper's interpretation is not that every connected executive causes fatalities. It is that connections lower the expected cost of safety noncompliance, leading to riskier behavior on average.

5 Tables and figures

5.1 Figure 1: Frequency distribution of deaths

Read: Figure 1 shows the distribution of nonzero firm-year worker deaths. Most positive-fatality observations are in the single digits, but the distribution has a right tail, concentrated in mining and construction.

Interpretation: The figure motivates why the paper uses several mortality measures: raw deaths, an indicator for any deaths, death rate per employee, and severe accidents. The right tail means that a few large events can affect raw-death regressions.

Economic message: Worker fatality is a rare but socially severe outcome. Studying both extensive-margin deaths and intensity measures makes the evidence more robust.

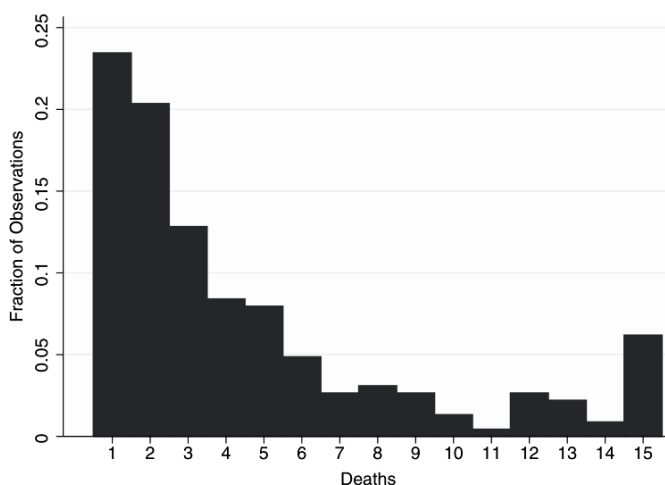


FIGURE 1

Frequency distribution of *Deaths*.

Notes: The sample used to construct this figure includes the 226 firm-year observations in which *Deaths* is positive. *Deaths* is firm-level worker fatalities per year. In this figure, the distribution of *Deaths* is censored at a maximum of 1

Figure 1: Frequency distribution of deaths.

5.2 Tables 1A and 1B: Summary statistics and connected versus unconnected firms

Read: Table 1A reports full-sample summary statistics. Connected equals 0.087, meaning about 8.7 percent of firm-years are politically connected. At least one fatality occurs in 13.5 percent of observations, and mean *DeathRate* is 0.029 per 1,000 employees. Table 1B shows connected firms have much higher log deaths, death indicators, death rates, and severe-accident rates, while also being larger and more profitable.

Interpretation: These tables establish the raw facts. Connected firms are not randomly comparable: they are larger, more state-owned, and more profitable. This motivates controls and fixed effects. But the mortality gap is already visible before regression adjustment.

Economic message: Political connections are associated with both private benefits and social costs: connected firms have higher ROA and higher worker mortality.

TABLE 1A
Summary statistics for firm-year panel data

Variable	Mean	SD	Median	Min	Max	Obs
$\log(1+Deaths)$	0.211	0.612	0.000	0.000	5.094	1475
$I(Deaths>0)$	0.135	0.342	0.000	0.000	1.000	1475
$DeathRate$	0.029	0.090	0.000	0.000	0.366	1475
<i>Severe</i>	0.039	0.194	0.000	0.000	1.000	1475
<i>Connected</i>	0.087	0.283	0.000	0.000	1.000	1475
<i>NSNP</i>	0.411	0.492	0.000	0.000	1.000	1475
<i>State Ownership</i>	0.180	0.246	0.000	0.000	0.971	1475
$\log(Sales)$	22.004	1.765	21.874	14.463	28.656	1475
$CapInt$	14.139	0.838	14.106	10.896	19.910	1475
<i>ROA</i>	0.039	0.066	-0.033	-0.221	0.220	1475
$\log(Employment)$	8.157	1.498	8.026	3.045	13.223	1475
<i>Investment</i>	0.078	0.065	0.065	-0.179	0.455	1475
<i>SafetyViol</i>	0.027	0.162	0.000	0.000	1.000	1475
<i>Pollute</i>	0.120	0.325	0.000	0.000	1.000	1388

Notes: All variables are at the firm-year level, and includes our full sample of 1475 firms. *Deaths* is firm-level worker fatalities per year; $I(Deaths>0)$ is an indicator variable denoting at least one fatality in a firm-year observation; $DeathRate$ is equal to $1000 \times (Deaths/Employment)$; *Severe* is an indicator variable denoting at least three fatalities from one accident in a firm-year observation; *Connected* is a firm-year indicator variable, equal to one if a senior executive at the firm previously served as mayor or vice-mayor in the city ("Tingju Ji" in Chinese) where the firm is located or as a provincial or central government official of the same or higher rank; *StateOwnership* is the ratio of state shares to total outstanding shares; $CapInt$ is log value of the ratio of total assets to total employees; *NSNP* is a province-year variable denoting whether a province had passed an NSNP law in a previous year; *SafetyViol* is an indicator variable denoting whether a firm was reported in the media for safety violations; *Pollute* indicates firm-year observations where the firm was fined for environmental violations.

(a) Table 1A: Summary statistics.

TABLE 1B
Summary statistics for firm-year panel data: politically connected versus politically unconnected

Variable	Politically Connected = 0	Politically Connected = 1	Difference	(Std Error)
$\log(1+Deaths)$	0.155	0.790	0.635***	(0.054)
$I(Deaths>0)$	0.110	0.395	0.285***	(0.031)
$DeathRate$	0.024	0.084	0.060***	(0.008)
<i>Severe</i>	0.027	0.163	0.135***	(0.018)
<i>NSNP</i>	0.417	0.349	-0.068	(0.045)
<i>State Ownership</i>	0.169	0.301	0.133***	(0.022)
$\log(Sales)$	21.871	23.388	1.517***	(0.158)
$CapInt$	14.132	14.211	0.079	(0.077)
<i>ROA</i>	0.036	0.067	0.031***	(0.006)
$\log(Employment)$	8.039	9.386	1.347***	(0.134)
<i>Investment</i>	0.077	0.091	0.014**	(0.006)
<i>SafetyViol</i>	0.022	0.078	0.055***	(0.015)
<i>Pollute</i>	0.119	0.123	0.004	(0.031)

Notes: All variables are at the firm-year level, and includes our full sample of 1475 firms. *Deaths* is firm-level worker fatalities per year; $I(Deaths>0)$ is an indicator variable denoting at least one fatality in a firm-year observation; $DeathRate$ is equal to $1000 \times (Deaths/Employment)$; *Severe* is an indicator variable denoting at least three fatalities from one accident in a firm-year observation; *Connected* is a firm-year indicator variable, equal to one if a senior executive at the firm previously served as mayor or vice-mayor in the city ("Tingju Ji" in Chinese) where the firm is located or as a provincial or central government official of the same or higher rank; *StateOwnership* is the ratio of state shares to total outstanding shares; $CapInt$ is log value of the ratio of total assets to total employees; *NSNP* is a province-year variable denoting whether a province had passed an NSNP law in a previous year; *SafetyViol* is an indicator variable denoting whether a firm was reported in the media for safety violations; *Pollute* indicates firm-year observations where the firm was fined for environmental violations.

*Significant at 10%; **significant at 5%; *** significant at 1%.

(b) Table 1B: Connected versus unconnected firms.

5.3 Table 1C and Figures 2A–2B: Within-firm variation and industry death rates

Read: Table 1C restricts to firms with variation in connectedness. The connected-unconnected mortality gap remains large in this more balanced sample. Figures 2A and 2B show that connected firms have higher death rates in most industries, in both the full sample and the within-firm-variation sample.

Interpretation: Table 1C and Figures 2A–2B are important because they show the mortality gap is not only a cross-sectional size or industry artifact. The gap persists when focusing on firms that change connectedness over time.

Economic message: Political connections are associated with worker deaths within comparable industry and firm-change contexts, not only across very different firms.

TABLE 1C
Firm-year panel data: politically connected versus politically unconnected firms, $Var(Connected)>0$ sample

Variable	Politically Connected = 0	Politically Connected = 1	Difference	(Std Error)
$\log(1+Deaths)$	0.345	0.732	0.387***	(0.104)
$I(Deaths>0)$	0.224	0.373	0.148***	(0.053)
$DeathRate$	0.039	0.099	0.060***	(0.014)
<i>Severe</i>	0.049	0.176	0.127***	(0.033)
<i>NSNP</i>	0.422	0.441	0.020	(0.059)
<i>State Ownership</i>	0.220	0.287	0.066**	(0.032)
$\log(Sales)$	22.629	22.772	0.143	(0.202)
$CapInt$	14.057	14.114	0.057	(0.116)
<i>ROA</i>	0.051	0.063	0.012*	(0.007)
$\log(Employment)$	8.772	8.864	0.092	(0.184)
<i>Investment</i>	0.090	0.085	-0.005	(0.008)
<i>SafetyViol</i>	0.072	0.059	-0.013	(0.030)
<i>Pollute</i>	0.171	0.071	-0.099**	(0.042)

Notes: All variables are at the firm-year level, and includes the sample of firms where there is variation in *Connected* during our sample period (the $Var(Connected)>0$ sample), a total of 325 observations. *Deaths* is firm-level worker fatalities per year; $I(Deaths>0)$ is an indicator variable denoting at least one fatality in a firm-year observation; $DeathRate$ is equal to $1000 \times (Deaths/Employment)$; *Severe* is an indicator variable denoting at least three fatalities from one accident in a firm-year observation; *Connected* is a firm-year indicator variable, equal to one if a senior executive at the firm previously served as mayor or vice-mayor in the city ("Tingju Ji" in Chinese) where the firm is located or as a provincial or central government official of the same or higher rank; *StateOwnership* is the ratio of state shares to total outstanding shares; $CapInt$ is log value of the ratio of total assets to total employees; *NSNP* is a province-year variable denoting whether a province had passed an NSNP law in a previous year; *SafetyViol* is an indicator variable denoting whether a firm was reported in the media for safety violations; *Pollute* indicates firm-year observations where the firm was fined for environmental violations.

*Significant at 10%; **significant at 5%; ***significant at 1%.

Table 1C: Connected versus unconnected firms in the $Var(Connected)>0$ sample.

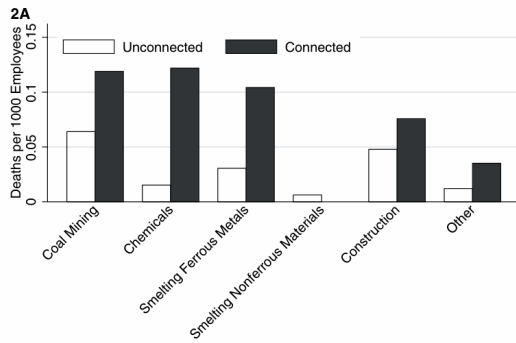


FIGURE 2A
Average DeathRate at Connected versus Unconnected Firms, by Industry (full sample).

(a) Figure 2A: DeathRate by industry, full sample.

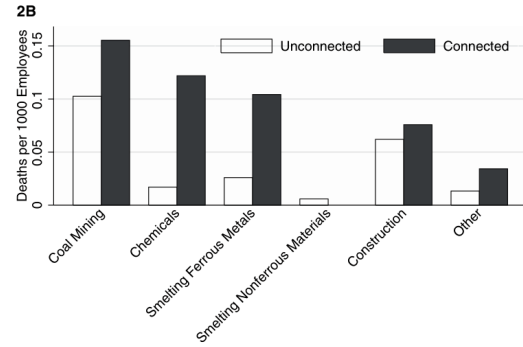


FIGURE 2B
Average DeathRate at Connected versus Unconnected Firms, by Industry (Var(Connected)>0 sample).
Notes: The figure is constructed using the Var(Connected)>0 sample of 325 firm-year observations. Var(C'd)>0

(b) Figure 2B: DeathRate by industry, Var(Connected)>0 sample.

5.4 Tables 2 and 3: Political connections and workplace deaths

Read: Table 2 estimates negative binomial regressions for annual deaths. The coefficient on Connected is positive and significant across specifications, including specifications with firm and year fixed effects. Table 3 estimates OLS models for log deaths, death-rate, and death indicators. Connected remains positive across many specifications, although precision varies in smaller within-firm samples.

Interpretation: These are the core fatality results. Table 2 is designed for count outcomes; Table 3 checks that the result is not an artifact of count-model assumptions. The firm fixed-effects specifications are particularly important because they rely on changes in executive connectedness within firms.

Economic message: Connectedness is associated with substantially higher worker mortality, suggesting that political ties can relax safety constraints.

TABLE 2
The impact of political connections on workplace deaths: negative binomial regression

	(1)	(2)	(3)	(4)	(5)	(6)
	Columns 1-6: Deaths					
Connected	1.953*** (0.393)	1.754*** (0.318)	1.697*** (0.328)	0.772*** (0.212)	1.521*** (0.398)	0.855*** (0.222)
log(Sales)		0.665*** (0.068)	0.725*** (0.096)	0.134* (0.074)	1.210*** (0.298)	0.329** (0.133)
ROA		4.626*** (1.675)	-1.259 (2.094)	0.267 (1.709)	1.514 (5.451)	2.731 (2.936)
CapInt		-0.256** (0.111)	0.093 (0.146)	-0.351** (0.150)	-0.066 (0.332)	-0.372* (0.216)
StateOwnership		1.671*** (0.429)	1.238** (0.588)	0.477 (0.368)	-0.401 (1.027)	0.680 (0.610)
Investment		0.304 (1.562)	0.519 (1.885)	-2.610 (1.720)	3.551 (3.957)	-2.216 (2.561)
Constant	-0.527*** (0.120)	-12.939*** (1.843)	-17.487*** (2.762)	0.210 (2.077)	-24.789*** (7.802)	-4.045 (3.446)
ln(α)						
Constant	2.881*** (0.090)	2.278*** (0.099)	1.786*** (0.103)		1.278*** (0.153)	
Fixed effects	-	-	Year, Industry; Province	Firm, Year	Year, Industry; Province	Firm, Year
Sample	Full	Full	Full	Full	Var(C'd)>0	Var(C'd)>0
Observations	1475	1475	1475	524	325	198
Log pseudolikelihood	-1095	-1014	-933	-506	-379	-223

Notes: All variables are at the firm-year level. Deaths is firm-level worker fatalities per year; Connected is a firm-year indicator variable, equal to one if a senior executive at the firm previously served as mayor or vice-mayor in the city ("Tingju Ji" in Chinese) where the firm is located or as a provincial or central government official of the same or higher rank; CapInt is log value of the ratio of total assets to total employees; Investment is the ratio of capital expenditure to total assets; StateOwnership is the ratio of state shares to total outstanding shares. Var(C'd)>0 denotes that we limit the sample to firms where there is variation in Connected over the sample period 2008-2013, due either to the arrival or departure of a connected executive. Robust standard errors are in parentheses; with the exception of columns 4 and 6 standard errors are clustered at the firm level.

*Significant at 10%; ** significant at 5%; *** significant at 1%.

(a) Table 2: Negative binomial regressions.

TABLE 3
The impact of political connections on workplace deaths: OLS regression

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
	Columns 1-3: log(1 + death)			Columns 4-6: DeathRate			Columns 7-9: I/Deaths > 0		
connected	0.282*** (0.106)	0.283** (0.138)	0.370** (0.150)	0.048*** (0.017)	0.050** (0.020)	0.063*** (0.021)	0.109** (0.049)	0.088 (0.056)	0.121** (0.060)
(Sales)	0.006*** (0.017)	0.013 (0.166)	-0.181 (0.157)	0.006** (0.002)	-0.009 (0.025)	-0.007* (0.024)	0.005*** (0.009)	-0.046 (0.082)	-0.009 (0.070)
ln	-0.038 (0.284)	-0.318 (0.360)	1.555 (1.220)	-0.015 (0.036)	-0.075 (0.108)	0.219 (0.155)	-0.035 (0.117)	-0.192 (0.358)	0.808* (0.457)
plnt	-0.064** (0.028)	0.112 (0.060)	0.158 (0.109)	0.001 (0.004)	0.019 (0.020)	0.031 (0.022)	-0.038** (0.016)	0.108 (0.068)	0.105** (0.050)
seOwnership	0.149* (0.083)	-0.007 (0.213)	0.363 (0.248)	0.013 (0.012)	-0.034 (0.026)	0.001 (0.034)	0.094** (0.047)	-0.009 (0.115)	0.239** (0.113)
vestment	0.074 (0.213)	-0.556 (0.621)	0.484 (0.639)	-0.007 (0.045)	-0.127 (0.105)	0.008 (0.117)	0.051 (0.153)	-0.344 (0.343)	0.251 (0.352)
nsaint	-0.739* (0.415)	2.249 (1.828)	1.941 (1.480)	-0.074 (0.072)	-0.044 (0.247)	0.117 (0.343)	-0.302 (0.230)	-1.301 (0.850)	0.106 (1.758)
Year, Industry; Province	Full	Full	Full	Year, Industry; Province	Full	Full	Year, Industry; Province	Full	Full
Year, Firm	Var(C'd)>0	Var(C'd)>0	Var(C'd)>0	Year, Industry; Province	Var(C'd)>0	Var(C'd)>0	Year, Industry; Province	Var(C'd)>0	Var(C'd)>0
Observations	1475	325	325	1475	325	325	1475	325	325
nsaint	0.28	0.36	0.493	0.138	0.322	0.45	0.268	0.406	0.512

Notes: All variables are at the firm-year level. Deaths is firm-level worker fatalities per year; I/Deaths>0 is an indicator variable denoting at least one fatality in a firm-year. Connected is equal to 1000*(Deaths/Employment). Connected is a firm-year indicator variable, equal to one if a senior executive at the firm previously served as mayor or vice-mayor in the city ("Tingju Ji" in Chinese) where the firm is located or as a provincial or central government official of the same or higher rank; CapInt is log value of the ratio of total assets to total employees; Investment is the ratio of capital expenditure to total assets; StateOwnership is the ratio of state shares to total outstanding shares. Var(C'd)>0 denotes we limit the sample to firms where there is variation in Connected over the sample period 2008-2013, due either to the arrival or departure of a connected executive. Robust standard errors, clustered at the firm level, are in parentheses.

(b) Table 3: OLS regressions.

5.5 Tables 4 and 5: Safety violations and environmental non-compliance

Read: Table 4 shows that in firm-years with no deaths, connected firms have zero public safety-violation reports, while comparable unconnected firms have a 4.6 percent report rate. In years with fatalities, violation rates are nearly identical for connected and unconnected firms. Table 5 shows connected firms are less likely to be fined for environmental non-compliance.

Interpretation: These tables support the ex ante evasion mechanism. Connected firms do not appear to be generally safer; rather, before fatal accidents make problems public, they face less visible enforcement.

Economic message: Political connections seem to reduce regulatory scrutiny across safety and environmental margins, consistent with lower effective compliance costs.

TABLE 4 Crosstab of Connected and I(Deaths>0) for average safety violation rate (SafetyViol)		
	Deaths = 0	Deaths > 0
Connected=0	0.046	0.160
Connected=1	0.000	0.158
Difference	0.046	0.002

Notes: Sample includes 325 firms with variation in *Connected* during 2008–2013; *Deaths* is firm-level worker fatalities per year; *I(Deaths>0)* is an indicator variable denoting at least one fatality in a firm-year observation; *Connected* is a firm-year indicator variable, equal to one if a senior executive at the firm previously served as mayor or vice-mayor in the city (“Tingju Ji” in Chinese) where the firm is located or as a provincial or central government official of the same or higher rank; *SafetyViol* is an indicator variable denoting whether a firm was reported in the media for safety violations.

(a) Table 4: Safety violations by connectedness and deaths.

TABLE 5 The impact of political connections on environmental non-compliance			
	(1)	(2)	(3)
	Columns 1–3: <i>Pollute</i>		
<i>Connected</i>	−0.099*** (0.036)	−0.093** (0.040)	−0.071* (0.038)
<i>log(Sales)</i>		0.031* (0.017)	0.007 (0.018)
<i>ROA</i>		0.143 (0.439)	−0.152 (0.276)
<i>Investment</i>		0.212 (0.332)	−0.130 (0.257)
<i>CapInt</i>		−0.016 (0.025)	0.021 (0.018)
<i>StateOwnership</i>		−0.115 (0.080)	0.199* (0.102)
<i>Constant</i>	0.171*** (0.040)	−0.315 (0.525)	−0.456 (0.522)
Fixed effects	—	—	Year; Industry; Province
Sample		Var(C’ d)>0	
Observations	315	315	315
R ²	0.0175	0.0418	0.398

Notes: All variables are at the firm-year level. *Pollute* indicates firm-year observations where the firm was fined for environmental violations; *Connected* is a firm-year indicator variable, equal to one if a senior executive at the firm previously served as mayor or vice-mayor in the city (“Tingju Ji” in Chinese) where the firm is located or as a provincial or central government official of the same or higher rank; *CapInt* is log value of the ratio of total assets to total employees; *Investment* is the ratio of capital expenditure to total assets; *StateOwnership* is the ratio of state shares to total outstanding shares. Var(C’ d)>0 denotes that we limit the sample to firms where there is variation in *Connected* over the sample period 2008–2013, due either to the arrival or departure of a connected executive. Robust standard errors, clustered at the firm level, are in parentheses.

*Significant at 10%; ** significant at 5%; *** significant at 1%.

(b) Table 5: Environmental non-compliance.

5.6 Table 6 and Figure 3: Firm performance and market reaction to fatal accidents

Read: Table 6 shows that connected firms are more profitable, but the profitability premium is lost after fatal accidents: the interaction $\text{Connected} \times \text{I(Deaths}>0)$ is negative in future ROA specifications. Figure 3 shows cumulative abnormal returns after fatal accidents. Connected firms suffer steadily declining abnormal returns over the 30 trading days after accidents, while unconnected firms do not.

Interpretation: Table 6 and Figure 3 connect the mortality mechanism to firm value. If connections generate profits by avoiding compliance, a fatal accident should reveal the fragility of that advantage and reduce value.

Economic message: Fatal accidents destroy the economic value of political protection. Investors price this loss at connected firms.

	(1)	(2)	(3)	(4)
	Columns 1-2: ROA		Columns 3-4: ROA_FI	
<i>Connected</i>	0.017*** (0.006)	0.011* (0.006)	0.015*** (0.006)	0.016** (0.007)
<i>log(Sales)</i>	0.004** (0.002)	0.007 (0.005)	0.001 (0.002)	0.002 (0.004)
<i>Investment</i>	0.146*** (0.035)	0.069 (0.074)	0.070** (0.030)	0.029 (0.044)
<i>CapInt</i>	0.008** (0.004)	0.008 (0.010)	0.006** (0.003)	0.007 (0.007)
<i>StateOwnership</i>	-0.011 (0.009)	-0.001 (0.027)	-0.003 (0.008)	-0.020 (0.017)
<i>I(Deaths > 0)</i>			0.005 (0.004)	0.005 (0.008)
<i>Connected * I(Deaths > 0)</i>			-0.017** (0.008)	-0.030*** (0.010)
<i>ROA</i>			0.358*** (0.057)	0.155 (0.095)
<i>Constant</i>	-0.106 (0.065)	-0.120 (0.160)	-0.050 (0.050)	0.004 (0.139)
Fixed effects		Year; Industry; Province		
Sample	Full	Var(Ci'd)>0	Full	Var(Ci'd)>0
Observations	1475	325	1199	268
R ²	0.259	0.477	0.362	0.601

Notes: All variables are at the firm-year level. *ROA_FI* refers to ROA in year $t+1$; *Deaths* is firm-level worker fatalities per year; *I(Deaths > 0)* is an indicator variable denoting at least one fatality in a firm-year observation; *Connected* is a firm-year indicator variable, equal to one if a senior executive at the firm previously served as mayor or vice-mayor in the city ("Tingju Ji" in Chinese) where the firm is located or as a provincial or central government official of the same or higher rank; *CapInt* is log value of the ratio of total assets to total employees; *Investment* is the ratio of capital expenditure to total assets; *StateOwnership* is the ratio of state shares to total outstanding shares. Var(Ci'd)>0 denotes that we limit the sample to firms where there is variation in *Connected* over the sample period 2008–2013, due either to the arrival or departure of a connected executive. Robust standard errors, clustered at the firm level, are in parentheses.
*Significant at 10%; ** significant at 5%; *** significant at 1%.

(a) Table 6: Political connections, deaths, and firm performance.

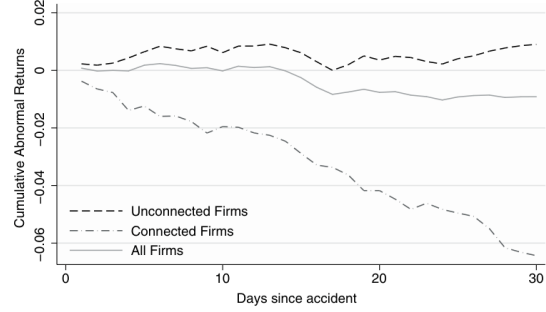


FIGURE 3
Market reaction to fatal accidents, Connected versus Unconnected firms.
Notes: Each data point represents the post-accident cumulative abnormal returns CARs[1,K], estimated using a standard una-French four-factor model, over the window of [1,K] where the accident date is set as 1. *Connected* is a firm-year indicator variable, equal to one if a senior executive at the firm previously served as mayor or vice-mayor in the city Tingju Ji" in Chinese) where the firm is located or as a provincial or central government official of the same or higher rank.

(b) Figure 3: Market reaction to fatal accidents.

5.7 Tables 7 and 8: Event-study regressions and loss of connections

Read: Table 7 confirms the market-reaction pattern in regressions. Connected firms experience negative CARs relative to unconnected firms, and the magnitude grows from short to long windows. Table 8 shows that firms are less likely to remain connected after fatal accidents, especially in the within-firm variation sample.

Interpretation: The two tables jointly show a plausible channel for value loss: after fatal accidents, connected executives leave or lose usefulness, and firm value falls. This supports the paper's claim that connections are valuable because they shield firms before accidents, not because they protect firms after visible accidents.

Economic message: The private value of political ties is contingent. Fatal accidents make those ties costly or ineffective, and the market reacts accordingly.

	(1)	(2)	(3)	(4)	(5)	(6)	(7)
	CARs[1,3]	CARs[1,5]	CARs[1,10]	CARs[1,15]	CARs[1,20]	CARs[1,25]	CARs[1,30]
<i>Connected</i>	-0.014** (0.007)	-0.027*** (0.010)	-0.034* (0.018)	-0.042** (0.018)	-0.048* (0.027)	-0.059* (0.032)	-0.087** (0.036)
<i>log(MonthlyDeaths)</i>	-0.004 (0.003)	-0.003 (0.004)	-0.003 (0.007)	-0.007 (0.007)	-0.012 (0.009)	-0.008 (0.011)	-0.004 (0.013)
<i>log(Sales)</i>	-0.002 (0.005)	-0.001 (0.007)	0.005 (0.007)	0.016** (0.007)	0.009 (0.011)	0.012 (0.014)	0.008 (0.014)
<i>ROA</i>	-0.059 (0.083)	-0.026 (0.107)	-0.147 (0.121)	-0.151 (0.128)	-0.185 (0.193)	-0.251 (0.225)	-0.196 (0.240)
<i>Constant</i>	0.075 (0.113)	0.037 (0.152)	-0.109 (0.171)	-0.358** (0.179)	-0.202 (0.271)	-0.279 (0.328)	-0.184 (0.350)
Fixed effects	Province, Industry and Year						
Observations	210	210	210	210	210	210	210
Adjusted R^2	0.0671	0.0946	0.0649	0.0635	0.0752	0.0362	0.062

Notes: A standard OLS model is used to estimate the impact of political connections on CARs. CARs[1,K] is the cumulative abnormal return, estimated using a standard Fama-French-Carhart four-factor model, over the window of [1,K] where the accident day is set as 1; *Connected* is a firm-year indicator variable, equal to one if a senior executive at the firm previously served as mayor or vice-mayor in the city ("Tingju Ji" in Chinese) where the firm is located or as a provincial or central government official of the same or higher rank; *log(MonthlyDeaths)* is the log value of the total deaths that take place within a month of each initial accident. Robust standard errors, clustered at the firm level, are in parentheses. * Significant at 10%; ** significant at 5%; *** significant at 1%.

(a) Table 7: Political connections and CARs.

	(1)	(2)	(3)
	Columns 1-3: <i>Connected_F1</i>		
<i>l(Deaths > 0)</i>	-0.073*** (0.024)	-0.248*** (0.064)	-0.244*** (0.078)
<i>Connected</i>	0.401*** (0.049)	0.103* (0.057)	0.046 (0.064)
<i>StateOwnership</i>	0.047 (0.041)	0.310** (0.147)	0.328 (0.210)
<i>log(Sales)</i>	0.009 (0.007)	-0.047 (0.030)	-0.177 (0.151)
<i>CapInt</i>	-0.008 (0.012)	-0.048 (0.039)	-0.157 (0.118)
<i>ROA</i>	0.128 (0.095)	0.566 (0.635)	0.194 (0.928)
<i>Investment</i>	0.137 (0.123)	-0.171 (0.507)	-0.285 (0.586)
<i>Constant</i>	0.037 (0.168)	1.978*** (0.731)	6.434** (2.939)
Fixed effects	Year, Industry, Province		
Sample	Full	Var(Ci'd)>0	Firm Var(Ci'd)>0
Observations	1199	268	268
R^2	0.351	0.244	0.332

Notes: All variables are at the firm-year level. *Connected* is a firm-year indicator variable, equal to one if a senior executive at the firm previously served as mayor or vice-mayor in the city ("Tingju Ji" in Chinese) where the firm is located or as a provincial or central government official of the same or higher rank; *Connected_F1* denotes *Connected* in year $t+1$; *Deaths* is firm-level worker fatalities per year; *l(Deaths > 0)* is an indicator variable denoting at least one fatality in a firm-year observation; *CapInt* is log value of the ratio of total assets to total employees; *Investment* is the ratio of capital expenditure to total assets; *StateOwnership* is the ratio of state shares to total outstanding shares. Var(Ci'd)>0 denotes that we limit the sample to firms where there is variation in *Connected* over the sample period 2008–2013, due either to the arrival or departure of a connected executive. Robust standard errors, clustered at the firm level, are in parentheses.

(b) Table 8: Worker deaths and loss of political connections.

5.8 Tables 9 and 10: NSNP laws and regulator incentives

Read: Table 9 shows that connected firms' death rate falls sharply after NSNP reforms in provinces that pass such laws. Table 10 shows this pattern in regressions: *Connected* is positive, while *Connected* $\times$ NSNP is negative and approximately offsets the direct connection effect in death-rate specifications.

Interpretation: These tables are the main institutional evidence. They suggest that when safety regulators have career incentives to meet safety targets, connected firms can no longer rely as effectively on protection from monitoring.

Economic message: Regulator incentives can mitigate the social cost of political connections. The problem is not only corrupt firms; it is also weak monitoring incentives.

	All Firms	Unconnected	Connected
<i>NSNP=0</i>	0.035	0.025	0.166
<i>NSNP=1</i>	0.018	0.017	0.027
<i>Difference</i>		0.008	0.139

Notes: Sample includes the 564 observations from firms headquartered in provinces that implemented an NSNP law during 2009–2012; *Deaths* is firm-level worker fatalities per year; *DeathRate* is equal to $1000 \times (\text{Deaths}/\text{Employment})$; *Connected* is a firm-year indicator variable, equal to one if a senior executive at the firm previously served as mayor or vice-mayor in the city ("Tingju Ji" in Chinese) where the firm is located or as a provincial or central government official of the same or higher rank; *NSNP* is a province-year variable denoting whether a province had passed an NSNP law in a previous year.

(a) Table 9: Connected and NSNP cross-tab.

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
	Columns 1-3: <i>DeathRate</i>			Columns 4-6: <i>ROA</i>			Columns 7-9: <i>Pollute</i>		
<i>connected</i>	0.079*** (0.024)	0.093*** (0.027)	0.104*** (0.028)	0.013* (0.007)	0.011 (0.008)	0.009 (0.010)	0.030 (0.049)	0.002 (0.048)	0.015 (0.056)
<i>NP</i>	0.000 (0.008)	0.012 (0.028)	0.010 (0.029)	-0.004 (0.006)	-0.005 (0.011)	-0.006 (0.013)	0.008 (0.031)	0.047 (0.079)	0.023 (0.083)
<i>connected * NSNP</i>	-0.087*** (0.029)	-0.113*** (0.033)	-0.110*** (0.036)	0.010 (0.012)	0.001 (0.011)	-0.001 (0.013)	-0.150** (0.066)	-0.189** (0.086)	-0.143 (0.091)
<i>(Sales)</i>	0.005*** (0.002)	0.007 (0.006)	-0.046* (0.024)	0.004** (0.002)	0.007 (0.005)	0.035 (0.024)	0.039*** (0.009)	0.006 (0.017)	0.016 (0.074)
<i>plnt</i>	0.001 (0.004)	0.000 (0.011)	0.027 (0.020)	0.008** (0.004)	0.008 (0.010)	-0.022 (0.010)	0.002 (0.014)	0.018 (0.017)	0.009 (0.035)
<i>seOwnership</i>	0.012 (0.012)	-0.052 (0.027)	0.001 (0.031)	-0.011 (0.009)	-0.001 (0.028)	-0.010 (0.025)	-0.063 (0.054)	0.190* (0.099)	0.039 (0.115)
<i>it</i>	-0.012 (0.035)	-0.082 (0.111)	0.309 (0.146)	-0.074 (0.045)	-0.085 (0.075)	-0.074 (0.081)	-0.141 (0.249)	-0.051 (0.281)	-0.051 (0.414)
<i>estment</i>	-0.007 (0.045)	-0.115 (0.101)	0.005 (0.123)	0.146*** (0.035)	0.068 (0.075)	0.022 (0.135)	0.085 (0.249)	-0.133 (0.249)	-0.049 (0.280)
<i>nature</i>	-0.085 (0.071)	0.010 (0.249)	0.686 (0.552)	-0.107* (0.065)	-0.116 (0.162)	-0.396 (0.496)	-0.761*** (0.279)	-0.421 (0.362)	-0.419 (1.443)
Fixed effects	Year, Industry, Province								
Sample	Full	Province Var(Ci'd)>0	Firm Var(Ci'd)>0	Full	Province Var(Ci'd)>0	Firm Var(Ci'd)>0	Full	Province Var(Ci'd)>0	Firm Var(Ci'd)>0
Observations	1475	325	325	1475	325	325	1368	315	315
	0.152	0.361	0.486	0.26	0.477	0.623	0.152	0.412	0.524

Notes: All variables are at the firm-year level. *Deaths* is firm-level worker fatalities per year; *DeathRate* is equal to $1000 \times (\text{Deaths}/\text{Employment})$; *Connected* is a firm-year indicator variable, equal to one if a senior executive at the firm previously served as mayor or vice-mayor in the city ("Tingju Ji" in Chinese) where the firm is located or as a provincial or central government official of the same or higher rank; *CapInt* is log value of the ratio of total assets to total employees; *Investment* is the ratio of capital expenditure to total assets; *seOwnership* is the ratio of state shares to total outstanding shares; *NSNP* is a province-year variable denoting whether a province had passed an NSNP law in a previous year.

(b) Table 10: Differential impacts of NSNP on firm-level outcomes.

5.9 Table 11 and Appendix Tables A.1–A.4: Robustness and institutional details

Read: Table 11 shows that connected firms have higher severe-accident rates, reducing the concern that the result is driven by hidden minor-fatality reporting. Table A.1 lists NSNP laws by province. Table A.2 reports propensity-score tests. Table A.3 adds industry-specific trends, and Table A.4 adds further NSNP robustness controls.

Interpretation: The appendix and robustness tables address under-reporting, selection, differential industry trends, and NSNP-specific confounds. They do not make the design fully experimental, but they make the interpretation more credible.

Economic message: The evidence remains consistent with a specific mechanism: political connections reduce compliance incentives, and stronger regulator incentives reduce the resulting worker deaths.

TABLE 11
The impact of connections on severe accident rates: a test of under-reporting

	(1)	(2)	(3)
Columns 1–3: Severe			
<i>Connected</i>	0.100*** (0.038)	0.144*** (0.052)	0.159*** (0.057)
<i>log(Sales)</i>	0.020*** (0.004)	0.040*** (0.012)	–0.032 (0.059)
<i>ROA</i>	0.027 (0.053)	–0.167 (0.235)	0.270 (0.361)
<i>Investment</i>	0.006 (0.072)	0.055 (0.281)	0.253 (0.350)
<i>CapInt</i>	–0.011 (0.008)	–0.026 (0.022)	0.019 (0.052)
<i>State Ownership</i>	–0.005 (0.025)	–0.043 (0.064)	–0.006 (0.081)
<i>Constant</i>	–0.232** (0.111)	–0.364 (0.463)	0.487 (1.519)
Fixed effects	Year; Industry; Province	Year; Industry; Province	Year; Firm
Sample	Full	Var(Ct'd)>0	Var(Ct'd)>0
Observations	1475	325	325
<i>R</i> ²	0.1	0.186	0.248

Notes: All variables are at the firm-year level. *Severe* is an indicator variable denoting at least three fatalities from one accident in a firm-year observation; *Connected* is a firm-year indicator variable, equal to one if a senior executive at the firm previously served as mayor or vice-mayor in the city (“Tingju Ji” in Chinese) where the firm is located or as a provincial or central government official of the same or higher rank; *CapInt* is log value of the ratio of total assets to total employees; *Investment* is the ratio of capital expenditure to total assets; *StateOwnership* is the ratio of state shares to total outstanding shares. Var(Ct'd)>0 denotes that we limit the sample to firms where there is variation in *Connected* over the sample period 2008–2013, due either to the arrival or departure of a connected executive. Robust standard errors, clustered at the firm level, are in parentheses.

*Significant at 10%; ** significant at 5%; *** significant at 1%.

Table 11: Severe accident rates and under-reporting test.

TABLE A.1
Instatement of NSNP laws by province

Province	Effective date	Passage date	Deaths	Firm-year obs	Population in 2005 (unit: 10,000)	GDP per capita in 2005 (unit: 1 RMB)
Guangdong	1 February 2005	1 February 2005	10	122	9194	24435.02
Heilongjiang	1 January 2007	28 Nov 2006	13	12	3820	14434.06
Guizhou	1 June 2006	1 June 2006	68	18	3730	5051.96
Ningxia	1 November 2006	27 September 2006	2	12	596	10239
Shanxi	29 January 2008	29 January 2008	246	63	3355	12495
Yunnan	18 September 2008	18 September 2008	19	51	4450	7835
Guangxi	1 January 2009	27 December 2008	5	24	4660	8787.729
Jilin	20 July 2009	8 January 2009	0	18	2716	13348
Hainan	2 June 2010	2 June 2010	0	6	828	10871
Jiangsu	23 November 2010	23 November 2010	15	104	7475	24560
Hebei	1 January 2011	8 November 2010	43	49	6851	14782.26
Henan	3 January 2011	3 January 2011	76	39	9380	11346.5
Hunan	22 March 2011	22 March 2011	2	54	6326	10426
Inner Mongolia	4 March 2011	4 March 2011	18	48	2386	16330.82
Sichuan	11 April 2011	11 April 2011	49	88	8212	9060
Chongqing	4 July 2011	4 July 2011	27	20	2798	10982
Jiangxi	3 May 2012	3 May 2012	7	18	4311	9440
Liaoning	1 February 2012	30 November 2011	23	54	4221	18983.2
Shandong	6 April 2012	6 April 2012	78	107	9248	20096.46
Shaanxi	4 January 2012	4 January 2012	0	29	3720	9899
Hubei	No safety quota	No safety quota	57	56	5710	11431
Qinghai	No safety quota	No safety quota	16	40	543	10044.74
Shanghai	No safety quota	No safety quota	43	57	1778	51474
Tianjin	No safety quota	No safety quota	2	6	1043	35783.19
Tibet	No safety quota	No safety quota	0	6	277	9114
Yunnan	No safety quota	No safety quota	19	43	2010	13108
Zhejiang	No safety quota	No safety quota	20	108	4898	27702.69
Anhui	No safety quota	No safety quota	31	74	6120	8675.145
Beijing	No safety quota	No safety quota	410	90	1538	45443.69
Fujian	No safety quota	No safety quota	26	35	3535	18645.84
Gansu	No safety quota	No safety quota	7	24	2594	7476.529

Notes: Since 2004, China has followed a “safety production quota” system, whereby the central government assigns a “death quota” to each province, and the provincial government allocates the quota among municipalities. In response, provinces began adopting NSNP policies that made promotion of safety regulators and other local government officials contingent on meeting the safety production target set for their region by the provincial government. Guangdong was the first province to adopt this policy in 2005, while others followed suit only in recent years. Deaths is the number of worker deaths among firms in our sample. Firm-year obs is the number of observations that firms based in each province contribute to our sample. Population is province population in 2005. GDP per capita is income per capita in 2005. Both population and income figures are obtained from CSMAR Regional Economy Database.

TABLE A.2
Propensity score tests

Dependent variable	Matching method	Number of treatment firms	Number of control firms	ATT	Std. Err.
<i>DeathRate</i>	Nearest-neighbor matching	129	91	0.292	0.129
<i>DeathRate</i>	Kernel matching	129	1165	0.326	0.131
<i>DeathRate</i>	Radius matching	123	1165	0.338	0.129

Notes: Stata codes come from Sascha Becker and Andrea Ichino (2002). ATT stands for “Average Treatment Effect on the Treated”. Bootstrap replication number is set at 100 in each case. A logit model is used in the propensity score test, where firms are matched on these dimensions: *log(Sales)*, *ROA*, *log(Employment)*, *Investment* and SIC-1 digit industry. The Epanechnikov kernel is used in the kernel matching where the bandwidth is set at 0.04, smaller than the default value of 0.06.

(a) Table A.1: NSNP laws by province.

(b) Table A.2: Propensity-score tests.

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
	Columns 1-3: log(1 + death)			Columns 4-6: DeathRate			Columns 7-9: kDeaths > 0		
<i>connected</i>	0.278*** (0.105)	0.283** (0.138)	0.361** (0.153)	0.047*** (0.016)	0.050** (0.020)	0.062*** (0.021)	0.108** (0.047)	0.090* (0.044)	0.120** (0.050)
<i>(Sales)</i>	0.102*** (0.018)	0.084 (0.167)	0.009 (0.151)	0.007*** (0.002)	0.001 (0.024)	-0.021 (0.020)	0.056*** (0.009)	-0.005 (0.079)	0.050 (0.066)
<i>U</i>	-0.024 (0.194)	-0.157 (0.917)	1.713 (1.254)	-0.014 (0.035)	-0.062 (0.100)	0.232 (0.151)	-0.050 (0.114)	-0.144 (0.307)	0.809* (0.469)
<i>plnt</i>	-0.062** (0.030)	0.072 (0.137)	0.150 (0.106)	0.002 (0.004)	0.014 (0.019)	0.032 (0.021)	-0.077** (0.016)	0.086 (0.065)	0.105* (0.056)
<i>seOwnership</i>	0.151** (0.085)	-0.084 (0.229)	0.207 (0.266)	0.011 (0.012)	-0.046 (0.029)	-0.021 (0.036)	0.070 (0.047)	-0.055 (0.117)	0.147 (0.110)
<i>constant</i>	-0.413 (0.433)	-2.236* (1.897)	-237.809 (166.483)	-0.054 (0.075)	-0.070 (0.254)	-45.332* (19.176)	-0.184 (0.232)	-1.363 (0.888)	-211.836*** (57.131)
<i>ed effects</i>	Year: Industry: Province Full	Year: Industry: Province Full	Year: Industry: Province Full	Year: Industry: Province Full	Year: Industry: Province Full	Year: Industry: Province Full	Year: Industry: Province Full	Year: Industry: Province Full	Year: Industry: Province Full
<i>servations</i>	1475	325	325	1475	325	325	1475	325	325
	0.307	0.4	0.525	0.157	0.36	0.482	0.294	0.461	0.563

Notes: All variables are at the firm-year level. All specifications include industry-specific time trends. *Deaths* is firm-level worker fatalities per year. *DeathRate* is equal to $Q^k(Deaths/Employee)$. *Connected* is a firm-year indicator variable, equal to one if a senior executive at the firm previously served as mayor or vice-mayor in the city ("Tingjun Li" (Hubei) where the firm is located or as a provincial or central government official of the same or higher rank. *CapInt* is log value of the ratio of total assets to total employees; *seOwnership* is the ratio of capital expenditure to total assets. *StateOwnership* is the ratio of state shares to total outstanding shares. *NSNP* is a province-year variable denoting whether a firm has passed an NSNP law in a previous year. *Police* indicates firm year observations where the firm was fined for environmental violations. *Var(CY'd)* > 0 denotes that we limit sample to firms where there is variation in *Connected* over the sample period 2008–2013, due either to the arrival or departure of a connected executive. Robust standard errors, clustered at the firm level, are in parentheses.

*** significant at 1%; ** significant at 5%; * significant at 10%.

(a) Table A.3: Industry-specific trends.

	(1)	(2)	(3)	(4)	(5)
	Columns 1-5: DeathRate				
<i>Connected</i>	0.131*** (0.044)	0.097*** (0.028)	0.098*** (0.032)	0.093*** (0.027)	0.772** (0.316)
<i>NSNP</i>	0.080 (0.052)	0.050 (0.043)	0.000 (.)	-0.033 (0.215)	0.017 (0.027)
<i>Connected * NSNP</i>	-0.157*** (0.049)	-0.132*** (0.036)	-0.151*** (0.048)	-0.114*** (0.033)	-0.127*** (0.034)
<i>log(Sales)</i>	0.012 (0.007)	0.008 (0.006)	0.010 (0.008)	0.006 (0.008)	0.006 (0.006)
<i>CapInt</i>	0.045** (0.017)	0.004 (0.011)	0.009 (0.014)	-0.000 (0.012)	0.002 (0.011)
<i>StateOwnership</i>	-0.082** (0.037)	-0.061** (0.026)	-0.055 (0.041)	-0.033 (0.027)	-0.034 (0.026)
<i>ROA</i>	-0.023 (0.128)	-0.199** (0.099)	-0.386* (0.203)	-0.082 (0.111)	-0.077 (0.110)
<i>Investment</i>	-0.133 (0.137)	-0.121 (0.118)	-0.191 (0.158)	-0.116 (0.102)	-0.113 (0.100)
<i>log(Sales) * NSNP</i>				0.002 (0.010)	
<i>Connected * log(GDP per Capita)</i>					-0.065** (0.030)
<i>Constant</i>	-0.764*** (0.246)	159.468*** (10.050)	-0.141 (0.323)	0.025 (0.289)	-0.006 (0.246)
<i>Fixed effects</i>	Year: Industry: Province	Year: Industry: Province	Year: Industry: Province X Year: Industry: Province	Year: Industry: Province	Year: Industry: Province
<i>Sample</i>	Var(CY'd) > 0; Var(NSNP) > 0	Var(CY'd) > 0; Var(CY'd) > 0	Var(CY'd) > 0	Var(CY'd) > 0	Var(CY'd) > 0
<i>Observations</i>	143	325	325	325	325
<i>R²</i>	0.38	0.46	0.668	0.361	0.379

(b) Table A.4: Additional NSNP robustness.

6 Interpretive synthesis

- **Main empirical pattern.** Connected firms have more worker deaths and severe accidents.
- **Regulatory mechanism.** Connected firms are less likely to be reported for safety and environmental violations before fatal accidents become visible.
- **Profit mechanism.** Connected firms are more profitable, but the profit premium weakens after fatal accidents.
- **Market validation.** Investors punish connected firms more after fatal accidents, consistent with a loss of valuable protection.
- **Personnel channel.** Connected executives are more likely to leave after fatal accidents, providing a direct channel through which connection value disappears.
- **Institutional solution.** NSNP laws reduce the excess mortality associated with connected firms, suggesting that regulator incentives can counteract political protection.

7 Short summary

- The paper studies publicly listed Chinese firms from 2008–2013.
- A firm is connected if a top executive previously held a high-ranking government office.
- Worker deaths are measured using official records, firm reports, construction-accident data, CSR reports, and media searches.
- Connected firms have two to three times higher worker death rates than unconnected firms.
- The relationship holds in firm fixed-effects specifications that use executive turnover.
- Connected firms have fewer public safety-violation reports in years without fatalities.
- Fatal accidents produce negative abnormal stock returns at connected firms and predict connected-executive departure.
- NSNP laws that incentivize safety regulators to meet targets attenuate the connected-firm death premium.
- The results suggest that political connections help firms avoid costly safety compliance rather than merely protecting them after accidents occur.

8 Conclusion

8.1 Core conclusion

Fisman and Wang show that political connections can have direct mortality costs. In their sample of publicly traded Chinese firms, politically connected firms experience substantially higher worker death rates, lower ex ante safety scrutiny, and higher profitability before fatal accidents. When accidents occur, the value of those connections declines.

8.2 Economic intuition

Connections reduce the expected cost of shirking on safety. If connected firms expect weaker monitoring, they can save on compliance costs and increase profits. But fatal accidents are visible and costly: they trigger attention from regulators and investors, reduce the value of connections, and cause connected executives to leave.

8.3 Incremental contribution in one sentence

The paper's distinctive contribution is to show that the private value of political connections can be purchased with worker lives, and that improved incentives for regulators can reduce this social cost.

8.4 Takeaway

Political connections are not only a distributional or rent-seeking issue. They can distort the enforcement of socially valuable regulation. The paper therefore connects political economy, corporate governance, labor safety, and regulatory design in a setting where the outcome is extreme and socially important: worker mortality.